FDME-RATE: Frequency-Domain-Based Multipath Extraction for Robust Acoustic TOA Estimation

Naizheng Jia, Weimeng Cui, Xinheng Wang, *Senior Member, IEEE*, Liu Yang, Can Xue, and Zhi Wang, *Member, IEEE*

***Abstract*—High-accuracy time of arrival (TOA) estimation is essential for indoor acoustic localization, but traditional TOA estimation methods often fail to obtain accurate and robust TOA estimation performance due to severe multipath challenges, such as peak extraction by general cross correlation (GCC). To address the problem, this article presents a novel acoustic TOA estimation technique called FDME-RATE, a high-accuracy acoustic ranging system. The proposed method utilizes GCC to determine the coarse signal arrival time, followed by an enhanced frequency modulated continuous wave (FMCW) technique to convert the various arrival paths into distinct frequencies within the spectrum. In addition, we employ an adaptive multiple signal classification (AMUSIC) algorithm to achieve high-resolution TOA estimation. Through simulations and experiments conducted in a reverberant corridor, the results indicate that our method achieves an average TOA ranging error of just 0.16 and 0.19 m at the 95% confidence level across a 30-m span. Within a 20-m range, there is only a centimeter-level error. Additionally, in non line-of-sight (NLOS) scenarios, our proposed method further enhances TOA-based ranging accuracy by an additional 0.3 m over the same distance. The experimental results also demonstrate that our method outperforms traditional methods and recent research in terms of both estimation accuracy and robustness.**

***Index Terms*—Acoustic, frequency modulated continuous wave (FMCW), general cross correlation (GCC), indoor localization, multipath, MUSIC, time of arrival (TOA) estimation.**

I. Introduction

RECENTLY, the rapid proliferation of smartphones and intelligent devices has significantly escalated the demand for accurate indoor localization. Indoor positioning systems (IPSs) based on smart sensors have become increasingly vital in enhancing the efficiency of Internet of Things (IoT) services. This technology plays a pivotal role in various sectors, including commerce, healthcare, disaster management, and industrial settings [1]. The marked increase in high-accuracy and low-cost location-based services (LBSs), accompanied by rising user engagement, offers significant opportunities to enhance IoT functionalities. Despite their potential, traditional localization systems, such as the global navigation satellite system (GNSS) and China's BeiDou/Compass [2] fall short in indoor environments, due to signal obstruction issues in covered areas like buildings and urban infrastructures [3]. To address these shortcomings, several alternative technologies have been proposed to replace GNSS, including WiFi [4], Bluetooth low energy (BLE) [5], ZigBee [1], ultrawideband (UWB) [6], and acoustic methods [7].

Among them, WiFi, particularly the round trip time (RTT) proposed in the 802.11.mc standard, is notably prevalent. However, WiFi positioning struggles to achieve sufficient accuracy, typically decimeters [8]. In existing WiFi-based localization methods, smart mobile devices consume quite a lot of power as WiFi interfaces need to be used for frequent access point (AP) scanning during the localization process [9]. Conversely, BLE, known for its low-energy consumption ($\leq$ 10 mW), offers indoor transmission within a range of 70–100 m in version 4.2 and up to 400 m in version 5.0. Nonetheless, under low-power mode, BLE's effective ranging distance reduces to about 10 m, suitable mainly for personal and household devices but at the cost of additional beacons. ZigBee is built upon the IEEE 802.15.4 standard concerned with the physical and MAC layers for low-power consumption and low cost [10]. A significant limitation arises from the incompatibility of most personal mobile devices with Zigbee technology, thereby restricting its application in indoor localization scenarios [7]. UWB, a low-power, high-accuracy tech with up to 1 Gb/s data rate for IPSs, operates in the 3.1–10.6-GHz spectrum, reducing band interference. Decawave [11] achieves a localization accuracy of up to 10 cm using time difference of arrival (TDOA) over distances reaching 300 m. Although precise, UWB incurs higher costs and is only compatible with a limited range of smartphones. At the same time, UWB can be affected easily by metallic and liquid surfaces [12], [13].

Acoustic signal-based localization can achieve high-localization accuracy by leveraging microphone sensors on smartphones at an affordable cost [14]. Additionally, due to the similarity between acoustic ranging and RF signals, various methods can be employed in acoustic signal-based localization. Acoustic localization approaches often utilize TDOA based on time stamps [15], [16], [17]. Therefore, accurate time of arrival (TOA) estimation is a crucial factor for ensuring indoor acoustic localization [18], [19]. The methodologies for

Manuscript received 28 June 2024; accepted 15 July 2024. Date of publication 22 July 2024; date of current version 9 October 2024. This work was supported by the National Key Research and Development Program of China under Grant 2021YFB3900800. *(Corresponding author: Zhi Wang.)*

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Digital Object Identifier 10.1109/JIOT.2024.3431953

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TOA estimation have been subjected to extensive discourse over the past few decades, and Chirp signals have become widely used. Reference [20], [21], [22] demonstrate the application of maximum general cross correlation (GCC) for TOA measurement. Reference [23] utilizes a robust peak candidate algorithm to avoid multipath with a room size of 10 m× 10 m. Reference [24] utilizes the method of signal power spectrum subtraction to suppress reverberation, but they can only get an accuracy of 30 cm in coverage of 30 m. Reference [25], [26] introduces a threshold to extract the first path. Reference [27] employs fractional Fourier transform (FrFT) iteration to obtain an estimation of TOA with an error of 30 cm at a distance of 30 m, with a lot of computing complexity. Reference [16] uses a two-stage algorithm to search the first path and remove the pseudo-peaks of correlation at a distance of 10–15 m.

Inspired by millimeter-wave radar technology, frequency modulated continuous wave (FMCW) has been adopted for precise and robust acoustic sensing. Reference [28], [29] demonstrate the utilization of FMCW for achieving millimeter-level ranging at short distances. However, these studies neglect the challenges posed by multipath and reverberation effects in scenarios exceeding 5 m. Reference [30] employs FMCW alongside the analysis of multipath frequency peaks for temperature measurement, yet it restricts its focus to a limited range of 20 cm to 30 cm, typical of smartphone length. In contrast, Mao et al. [31] investigated the application of MUSIC in the frequency domain for distance estimation, and [32] combines FMCW with MUSIC for mutual mobile phone distance measurements up to 5 m. Despite their contributions, these studies do not extensively address multipath effects, MUSIC parameters, or compare with alternative methodologies. Moreover, they are limited to a 5-m distance range and do not tackle non line-of-sight (NLOS) ranging challenges. Furthermore, within intricate multipath settings, the traditional MUSIC algorithm frequently encounters spurious peaks due to uncertainties in the number of signal sources, resulting in a significant reduction in detection accuracy. Consequently, there is a pressing need for enhancements to the conventional MUSIC algorithm.

In practical scenarios, achieving high accuracy and robustness in acoustic TOA estimation presents several challenges, outlined as follows.

1) *Multipath Interference:* Acoustic signal propagation is notably prone to environmental reflections and multipath interference, such as those caused by walls, furniture, and ceilings. The GCC method struggles to differentiate signal paths in cluttered environments, complicating the identification process in multipath conditions.
2) *Long-Distance Detection:* The attenuation of high-frequency sound signals over long distances is considerable, thereby complicating their detection. The GCC method faces difficulties in distinguishing between noise and long-distance multipath interference. Moreover, there is a lack of research focusing on the application of FMCW technology for acoustic ranging in scenarios extending beyond 5 m.
3) *NLOS Propagation:* Obstructions in indoor mobile phone positioning, particularly those caused by handheld devices, pose significant challenges. Occlusions not only generate multipath conditions but also introduce substantial errors due to the inability to detect direct signal paths, further complicating accurate positioning.

These challenges underscore the complexity of developing effective acoustic TOA estimation methods suitable for diverse operational environments. To overcome the challenges, we propose a novel method to realize robust acoustic TOA estimation based on multipath extraction in a low-cost and commercial device. Our main contributions are shown as follows.

1) We propose a novel method for estimating the TOA of acoustic signals with high precision over long distances in the indoor environment, utilizing low-cost devices. This technique, which relies on enhanced FMCW and adaptive multiple signal classification (AMUSIC) algorithms, effectively converts multipath signals into the frequency domain. Compared to the traditional MUSIC algorithm, AMUSIC employs a method of adaptive parameter adjustment, effectively addressing the issue of high-resolution identification of multipath signals in complex environments with uncertain channel models. This transformation allows for high-resolution analysis, which is pivotal is pivotal for the exclusion of nondirect path signals, culminating in a TOA estimation of considerable accuracy.
2) We conducted a comprehensive comparative analysis between our proposed method and existing techniques through simulations and experimental validation. The results represent the best performance in the current state-of-the-art.
3) We performed experiments in LOS scenarios across various distances. The experimental results show that our proposed method can obtain an accuracy of 0.034 m within a 5-m range and 0.16 m within a 30-m range. In NLOS, it can also obtain a maximum error of 0.25 m within a 30-m range.

The structure of this article is outlined as follows: Section II delineates the FDME-RATE algorithm and provides an exposition of its theoretical underpinnings. Section III presents the outcomes of numerical simulations and ray acoustic simulations, including a comparative analysis with ablation simulations and other established methods. Section IV details the experimental results obtained from real-world implementations, alongside comparisons with other reference methodologies. Finally, Section V summarizes the conclusions of this article.

II. Methodology

Fig. 1 presents the comprehensive framework of the TOA estimation method proposed in this study. Initially, a bandpass filter is applied to eliminate noise, followed by the utilization of a matched filter to ascertain the preliminary TOA value corresponding to the maximum peak. Subsequently, the improved FMCW technique, developed in this article, is employed to transform the various arrival paths of the signals into frequency spectrum, facilitating the identification of the

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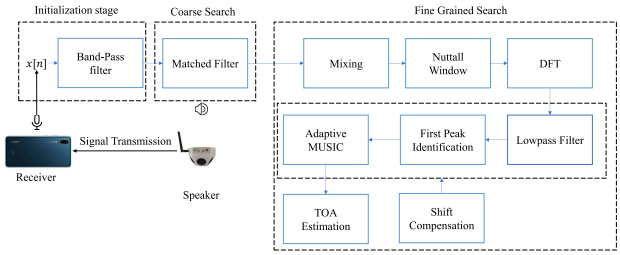


Fig. 1. Flowchart of the proposed method.

first path of TOA. Finally, AMUSIC algorithm is implemented to dynamically adjust the parameters of the MUSIC algorithm, enabling high-resolution spectral analysis and thus achieving precise TOA estimation. In practical implementations, almost all smartphone speakers are capable of emitting and receiving sound at a sampling rate of 48 kHz, thereby offering a foundational basis for cost-effective acoustic TOA estimation and localization techniques. In this article, we utilize a Chirp signal as the transmitted signal for indoor TOA estimation. The Chirp signal is depicted below

$$s(t) = \cos\left(2\pi f_{\min} t + \frac{\pi B t^2}{T}\right) \tag{1}$$

where $f_{\min}$ represents the initial frequency, B denotes the band frequency, and T represents the duration of the signal. The frequency bandwidth that smartphones can receive is limited to the range of 20 Hz to 24 kHz. However, according to [13], the lower threshold of human auditory perception for inaudible sounds extends beyond 17 kHz. Consequently, we may select $f_{\min}$ as 17 kHz.

When a loudspeaker emits a signal, spectrum leakage occurs during the initiation and termination phases, rendering the signal audible. To solve the problem, we invoke the Blackman window to make the transmit signal temporally smoother. Supposing N is the length of the window, Blackman window is formulated as:

$$w(n) = 0.42 - 0.5\cos\left(\frac{2\pi n}{N-1}\right) + 0.08\cos\left(\frac{4\pi n}{N-1}\right). \tag{2}$$

In a noiseless and reflection-free ideal environment, the received signal can be expressed as follows:

$$r(t) = \cos\left(2\pi f_{\min}(t-\tau_d) + \frac{\pi B(t-\tau_d)^2}{T}\right) \tag{3}$$

where τ_d is the signal propagation time delay and $\tau_d = d/c$. The sound velocity is c, and the distance of signal transmission is d. Fig. 2(a) depicts the ideal signal waveform, characterized by its clarity and absence of reverberation effects.

However, in practice, the received signal typically consists of a direct signal and several reflected signals due to the presence of multipath propagation. Fig. 2(b) illustrates the practical signal. The receiving signal, impacted by L paths interference, is depicted as follows:

$$s_r(t) = \sum_{i=1}^{L} a_i \cos\left(2\pi f_{\min}(t-\tau_i) + \frac{\pi B(t-\tau_i)^2}{T}\right) \tag{4}$$

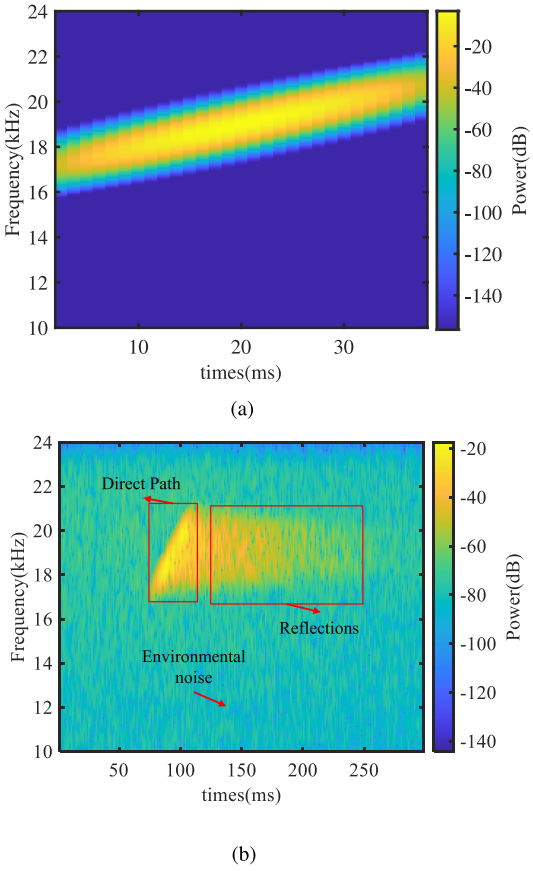


Fig. 2. Signal spectrum in the ideal (a) and practical (b) environment.

where a_i is the signal attenuation and τ_i is the time-of-flight of different paths. From (4), the receiving signal can be thought of as a series of superimposed signals with different time delays.

The conventional approach employs GCC to achieve time delay estimation. The computation of GCC can be expressed as [27]

$$R_{sr}(t) = \sum_{i=1}^{L} a_i R_{ss}(t)\delta(t-\tau_i) + R_{n_{ic}s}(t) \tag{5}$$

where R_{ss} represents the auto-correlation of the transmitting signal and $R_{n_{ic}}s$ denotes the cross-correlation between the noise and the transmitting signal. It can be found that the precision of $R_{sr}(t)$ is decided on R_{ss}. R_{ss} of Chirp can be described as

$$|R_{ss}| = \frac{\sqrt{C\left(\frac{\pi}{2}BT\right)^2 + S\left(\frac{\pi}{2}BT\right)^2}}{\sqrt{BT}} \tag{6}$$

where C and S are Fresnel parameters. Equation (6) shows that the value of R_{ss} is closely linked to its bandwidth and duration, with a wider bandwidth and longer duration typically resulting in larger R_{ss}.

The signal propagation time delay is obtained by GCC as

$$\tau_0 = \arg\max_t \{R_{sr}(t)\}. \tag{7}$$

However, based on (5), in environments with reverberation, which are impacted by multipath effects, (7) fails to yield

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accurate TOA estimations. Hence, we propose the use of FMCW signals, which enable direct separation of multipath and extraction of the desired direct signal.

Unlike traditional approaches that utilize long bandwidth signals or encoded acoustic signals, FMCW can convert Chirp signals with different time delays to single frequency signals by using $\cos A \cos B = ([\cos(A+B)+\cos(A-B)]/2)$. Suppose that we have an ideal receiving signal and the TOA is τ_d, the signal received can be multiplied with the original signal to generate a mixed single frequency signal, which is shown as

$$\begin{aligned} s_{\text{mix}}(t) &= s(t)r(t) \\ &= \frac{1}{2}\cos\left(2\pi f_{\min}\tau_d + \pi B\left(\frac{2t\tau_d - (\tau_d)^2}{T}\right)\right) + \text{high freq.} \\ &= \frac{1}{2}\cos\left(2\pi f_p t + \phi\right) + \text{high freq.} \end{aligned} \tag{8}$$

where $\phi = 2\pi f_{\min}(d/c)$ and $f_p = [(Bd)/(cT)]$. It can be concluded that the mixture consists of two components: 1) a low-frequency signal and 2) a high-frequency component. As the frequency of the high-frequency component (high freq.) is much higher than that of the low-frequency component, a low-pass filter can be utilized to eliminate the high-frequency interference.

By detecting the frequency component of the mixed signal, i.e. f_p, the TOA can be calculated as

$$d = \frac{f_p cT}{B}. \tag{9}$$

Next, we consider the received signal in practice. Substituting (4) into the expression, the mixed signal (intermediate frequency signal) after filtering can be represented as

$$\begin{aligned} x_{\text{mix}} &= s(t)s_r(t) \\ &= \frac{1}{2}\sum_{i=1}^{L} a_i \cos\left(2\pi f\tau_i + \pi B\left(\frac{2t\tau_i - (\tau_i)^2}{T}\right)\right) \\ &\approx \frac{1}{2}\sum_{i=1}^{L} a_i \cos(2\pi(f_{\min}\tau_i + B\tau_i t/T)). \end{aligned} \tag{10}$$

Upon conducting a fast Fourier transform (FFT) analysis, the distinct paths of the signal can be segregated in the frequency domain, as illustrated by

$$\begin{aligned} X_{\text{mix}}(\omega) &= \mathcal{F}\left(\sum_{i=1}^{L} a_i \cos(2\pi(f_{\min}\tau_i + B\tau_i t/T))\right) \\ &= \sum_{i=1}^{L} a_i \pi \delta(\omega + 2\pi B\tau_i/T). \end{aligned} \tag{11}$$

Obtaining f_p can be achieved by performing FFT. Nonetheless, spectral leakage present in the signal necessitates the application of a window function to alleviate its effects. To avoid interference between signals of different frequencies, the partials of the signal window must decay at a sufficiently rapid rate. In this regard, a 4-item and 5-order Nuttall window is employed due to its low-sidelobe level (−60.95 dB) and

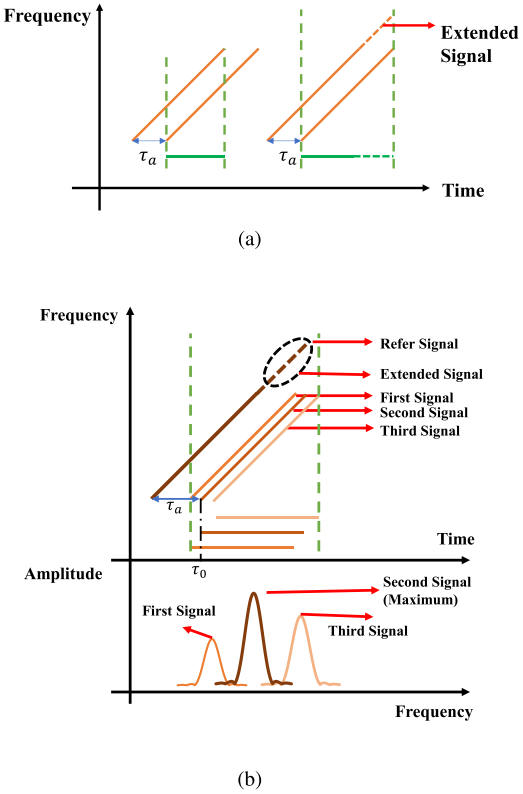


Fig. 3. Schematic representation of the multipath handling method. (a) Illustrates the extension of the reference signal. (b) Demonstrates the conversion of multipath signals into the frequency domain, where the second path is considered the maximum path.

decay rate (42 dB/oct) [33]. The Nuttall window function is defined as follows:

$$w_{\text{Nuttall}}(n) = \sum_{i=0}^{M-1} (-1)^i b_m \cos(2\pi ni/N) \tag{12}$$

where N is the length of the signal, $n = 0, 1, \ldots, N-1$ and M is the item number of the Nuttall, and b_m is the coefficient.

It can be observed from (13) that as long as B/T remains constant in the original signal, the measured distance will remain unchanged. Therefore, Bandwidth and signal duration can be extended in equal proportions. As Fig. 3(a), the effective length of the signal can be extended and the frequency calculation will be more accurate.

Based on (4), it is often the case that the maximum peak coincides with either the first path (where the impact of multipath is not significant) or lies beyond the first path (where the distance is greater or the impact of multipath is significant). In above situations, it is necessary to search for the TOA before the maximum peak. To address this issue, we introduce an advanced time parameter, denoted as $\tau_a = 10$ ms (generally the reflection path cannot be far away from 3 m, and in air, $0.01c = 3.43$ m), which is set before the maximum peak. Fig. 3(b) elucidates the procedure for transforming the multipath signal into the frequency domain. Also, based on (13), the cutoff frequency of the low-pass filter is shown as follows:

$$f_{\text{filter}} = \frac{\tau_a B}{T}. \tag{13}$$

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After that, the proposed method to estimate $\hat{f}_0$ is shown as

$$\hat{f}_0 = \frac{\underset{\omega}{\text{Extremum}}\{\text{peaks}[X_{\text{mix}}(\omega)]\}}{2\pi} \tag{14}$$

where Extremum is extremal operator and peaks is peak operator. By utilizing a threshold that is linked to the maximum spectral amplitude, it is possible to estimate the mixed signal frequency of the first path component

$$\hat{f}_0 = \frac{\min_{\omega}\{X_{\text{mix}}(\omega) \geq \lambda \max[X_{\text{mix}}(\omega)]\}}{2\pi} \tag{15}$$

where λ is a numerical factor used to determine the size of the threshold. The selection of the parameter λ has a direct impact on the detection accuracy and stability of the first path component. Empirical investigations will be conducted to identify the optimal value of λ.

The final TOA d_f estimation is depicted as

$$d_f = c\tau_0 - c\tau_a + \hat{f}_0 cT/B. \tag{16}$$

Nevertheless, the accuracy of TOA estimation by this method is constrained due to the frequency estimation resolution limitation, given by $\Delta f = 1/T$. Consequently, as derived from (16), this translates into the following resolution of distance estimation:

$$\Delta d_f = \frac{c}{B} = 8.575\text{cm}. \tag{17}$$

Equation (17) demonstrates that the current frequency resolution fails to distinguish multipath components below 8.575 cm. In environments with negligible multipath interference, this method achieves optimal detection accuracy. However, in contexts marked by multiple propagation paths and pronounced reverberation, real-time distance measurement is impractical. Furthermore, although window functions are employed to reduce sidelobe effects, these sidelobes persistently affect threshold determination. Consequently, we employ the MUSIC algorithm. Initially, the frequency vector **Feq** is defined as follows:

$$Feq = \left[Feq_{low}, Feq_{low} + Feq_{res}, \ldots, Feq_{low} + kFeq_{res}\right] \tag{18}$$

where Feq_{res} is the resolution of frequency, Feq_{low} and Feq_{high} are the bound of **Feq**, and $k = \lfloor(\text{Feq}_{\text{high}} - \text{Feq}_{\text{low}})/\text{Feq}_{\text{res}}\rfloor$.

For the received signal, denoted as $s_r(t)$, the corresponding signal matrix is represented by X_mix is defined as follows:

$$\mathbf{X}_{\text{mix}} = \left[x_{\text{mix}}(t) \quad \mathbf{D}x_{\text{mix}}(t) \cdots \mathbf{D}^{N-R+1}x_{\text{mix}}(t)\right] \tag{19}$$

where N represents the total number of samples of the signal $x_{\text{mix}}(t)$, R denotes the matrix order, and **D** is a 0/1 permutation matrix inducing a single-sample delay, defined as

$$\mathbf{D} = \left\{d_{ij}\right\}_{i,j=1,\ldots,N}, \quad d_{ij} = \begin{cases} 1, & i+1=j \\ 0, & \text{otherwise.} \end{cases} \tag{20}$$

The sample matrix **X** is shown as

$$\mathbf{X} = \mathbf{B}\mathbf{X}_{\text{mix}} \tag{21}$$

where **B** is an incidence matrix selecting only the last R rows of the matrix

$$\mathbf{B} = \{b_{ij}\}_{i=1,\ldots,R,j=1,\ldots,N}, b_{ij} = \begin{cases} 1, & i=j \\ 0 & \text{otherwise.} \end{cases} \tag{22}$$

The covariance matrix can be calculated as follows:

$$\mathbf{R} = \mathbf{X}\mathbf{X}^T/(N-R+1). \tag{23}$$

The singular value decomposition (SVD) composition of **R** is shown as

$$\mathbf{R} = \mathbf{U}\Sigma\mathbf{U}^T. \tag{24}$$

Here, **U** represents an orthogonal matrix comprised of the left singular vectors of **R**. The matrix Σ is a diagonal matrix, encapsulating the singular values of **R**. Additionally, $\mathbf{U}^T$ denotes the transpose of **V**, an orthogonal matrix whose columns are the right singular vectors of **R**.

We select the first P largest eigenvalues from the matrix Σ and simultaneously remove the corresponding rows of the matrix **U**, forming the noise subspace matrix $\mathbf{U_n}$ with the remaining elements.

The frequency matrix is shown as

$$\mathbf{G} = \exp\{-2\pi i\mathbf{Feq}\mathbf{M}\} \tag{25}$$

where **M** is shown as

$$\mathbf{M} = \begin{bmatrix}0 & 1 & 2 & \cdots & R\end{bmatrix}. \tag{26}$$

For each column θ of the matrix **G**, iterating through yields the following result:

$$\mathbf{P}(\theta)_{\text{MUSIC}} = \frac{1}{G(\theta)^H G(\theta)\mathbf{U_n}\mathbf{U_n}^H}. \tag{27}$$

The final frequency $\hat{f}_{\text{MUSIC}}$ is shown as

$$\hat{f}_{\text{MUSIC}} = \min_{\theta}\left\{\mathbf{P}(\theta)_{\text{MUSIC}} \geq \lambda_2 \max\left[\mathbf{P}(\theta)_{\text{MUSIC}}\right]\right\}. \tag{28}$$

Subsequently, by inserting $\hat{f}_{\text{MUSIC}}$ into (16), the revised TOA value is computed. Usually, due to the low-sidelobe characteristic of AMUSIC, the parameter λ_2 can be set to a minimal value, such as 0.2. The optimal parameters of λ and λ_2 are shown in Section IV.

Fig. 4 offers a detailed comparative analysis of the GCC and our proposed method applied to the same received signal. As evident from Fig. 4(a), the GCC method faces significant challenges in precisely identifying the arrival path in environments with dense reverberation. This difficulty primarily derives from the intensive arrival paths. Additionally, the presence of noise further complicates the accurate identification of the first path. In Fig. 4(a), it is observed that within the candidate region potentially representing the first path, numerous minor peaks are present, which are often challenging to differentiate. The process of extracting the signal's envelope leads to the omission of critical peak information, resulting in the conflation of the first and maximal paths, thereby significantly increasing ranging errors. In contrast, as depicted in Fig. 4(b), the proposed method discerns the precise arrival path with relative ease. Furthermore, Fig. 4(b) demonstrates that the proposed method not only enhances resolution but also effectively minimizes sidelobe interference.

However, the effectiveness of the MUSIC algorithm critically relies on its parameters, particularly the number of sources P. As illustrated in Fig. 5(a) and (b), an incorrect P value can result in the elimination of the correct path. Notably,

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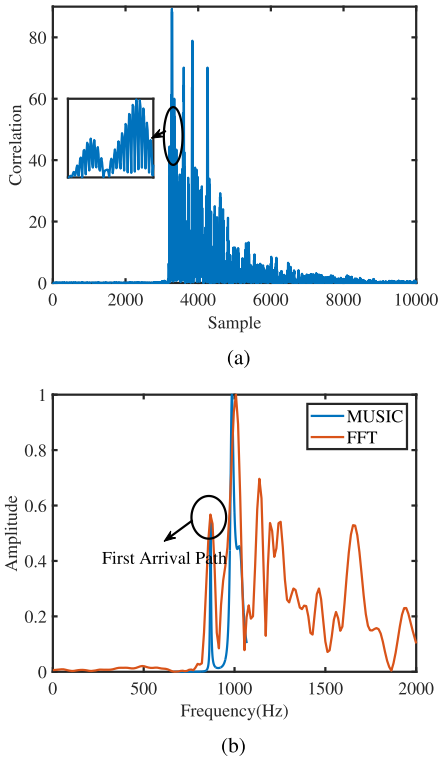


Fig. 4. Example of GCC and proposed method output. FFT represents the spectrum analysis of the multipath converse by FMCW. MUSIC represents the output of MUSIC algorithm refine the output. (a) GCC output. (b) Proposed method.

existing studies like [32] do not provide a definitive method for accurate parameter determination. Fig. 5(a) presents the spectral analysis conducted using the conventional MUSIC method for initial path detection with a small parameter P will cause the correct spectral peak to disappear. Conversely, Fig. 5(b) indicates that an overly large estimated P-value can lead to spurious spectral peaks, manifesting as two false spectral lines adjacent to the actual peak. To overcome this problem, we propose an adaptive MUSIC algorithm, outlined in Algorithm 1, designed to refine parameter accuracy. This algorithm fundamentally employs MUSIC to iteratively analyze and compare the positions and numbers of FFT peaks, thereby determining the optimal number of MUSIC signal sources.

Fig. 5(c) validates the efficacy of the AMUSIC algorithm in eliminating interference and attaining precise frequency estimation. It is observed that AMUSIC effectively addresses the challenges posed by the parameter setting of P in the traditional MUSIC method, accurately delineating the position of the first path and enhancing its resolution. The application of AMUSIC significantly diminishes the TOA estimation error, specifically by approximately 0.2 m.

III. Simulation

In this section, we present a detailed simulation framework to demonstrate the efficacy of our proposed method, which includes both numerical simulations and indoor ray acoustic simulation (IRAS). The numerical simulations aim to establish

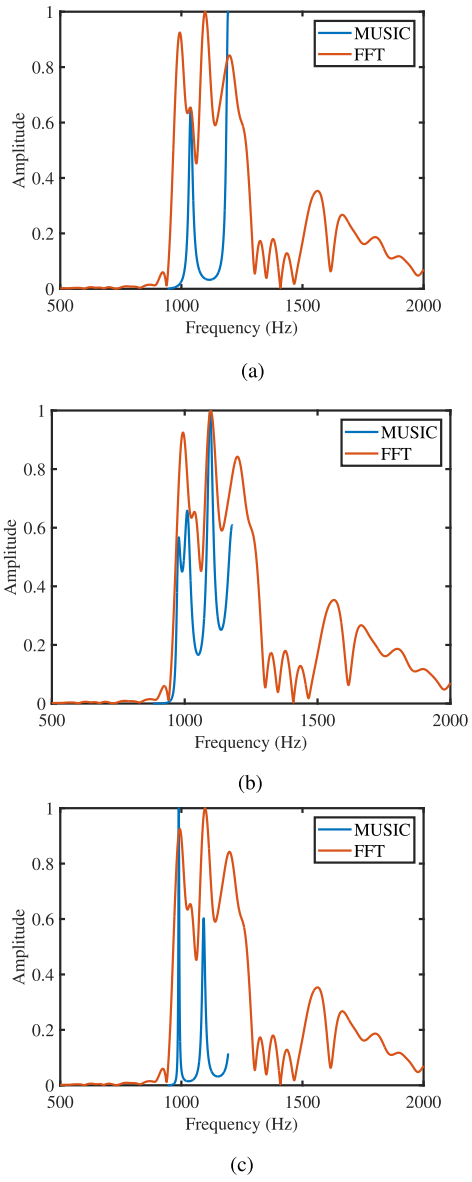


Fig. 5. Comparative outputs of FMCW, MUSIC, and AMUSIC. (a) Spectrum FMCW (FFT) and MUSIC (too small P). (b) Spectrum with FMCW (FFT) and MUSIC (too large P). (c) Spectrum with FMCW (FFT) and AMUSIC.

the viability of our approach within a completely random channel response model, including an ablation experiment, to validate our algorithm. Concurrently, IRAS corroborates this viability and evaluates the performance of alternative methods within the simulated actual sound field.

A. Numerical Simulation

In indoor environments, the channel impulse response (CIR) is defined as [24]

$$h(t) = \begin{cases} g(t) \cdot e^{-\Delta t}, & t \geq 0 \\ 0, & t < 0 \end{cases} \tag{29}$$

where $g(t)$ represents zero-mean Gaussian white noise and Δ is a factor associated with the reverberation time T_r, given by

$$\Delta = \frac{3 \ln 10}{T_r}. \tag{30}$$

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Algorithm 1 Adaptive MUSIC Algorithm

Input: Initial frequency estimate $\hat{f}_0$, mixed-signal X_{mix}, number of sources P
Output: Estimated frequencies $\hat{f}_{MUSIC}$

```
Define MUSIC Search Space: θ ∈ [f̂0 − 200, f̂0 + 200]
Compute MUSIC spectrum P_MUSIC
Identify peaks in MUSIC spectrum: Θ_MUSIC = peaks(P_MUSIC) {Based on (28)}
Identify FFT peaks and widths: Θ_FFT = peaks(X_mix), with corresponding widths w
while Number of peaks in the interval Θ_FFT[1] < Θ_MUSIC < Θ_FFT[2] is less than 2 do
    Increment P by 4
    Recompute MUSIC spectrum P_MUSIC
end while
Redetermine peaks in MUSIC spectrum: Θ_MUSIC = peaks(P_MUSIC)
while Θ_MUSIC[1] > Θ_FFT[2] + w[2]/2 OR Θ_FFT[1] < Θ_FFT[1] + w[1]/2 do
    Increment P by 4
    Recompute MUSIC spectrum P_MUSIC
end while
Final peaks detection: (28)
return f̂_MUSIC
```

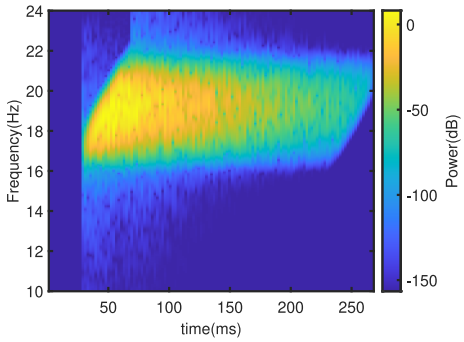


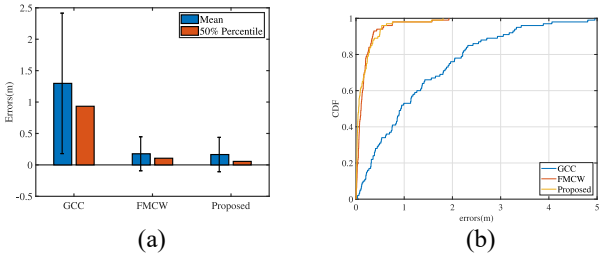
Fig. 6. Numerical simulation of one received signal spectrum, with reverberation.

The received signal in numerical simulations is described by

$$r(t)_{NM} = s(t) * h(t) + w(t) \tag{31}$$

where * denotes the convolution operation, and $w(t)$ is the Gaussian white noise of the environment. In Section III-A, we focus primarily on the multipath effects within the environment, rather than noise. The distance for the simulation is set at 10 m. Fig. 6 shows that the simulated signal, affected by reverberation, closely mirrors the signal in real-world environments.

Fig. 7 presents an ablation comparison TOA performance in terms of mean errors and 50% percentile errors observed in 100 Monte Carlo simulations using different methods. Here, GCC refers to the approach that considers the TOA based on the maximum value of the MF. The FFT method relies only on the frequency domain to identify the first path [19] without AMUSIC. The comparative analysis underscores the superior accuracy of our method over others. Specifically, the GCC method exhibited mean and 50th percentile errors of 1.29 and 0.9331 m, respectively. In contrast, the FFT method showed improved accuracy with errors of 0.17 and 0.106 m,



(a) (b)

Fig. 7. Numerical simulation comparative results. The bar charts in blue are the mean errors of the distance ranging. The bar charts in orange are the 50th percentile of errors. The error bar is the standard error. (a) Comparison bar chart of numerical simulation errors. (b) Numerical simulation CDF diagram.

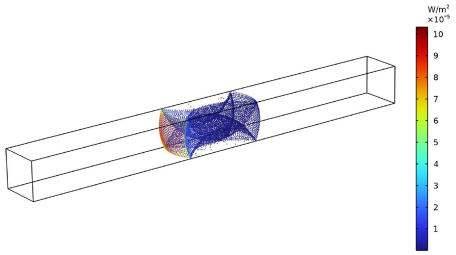


Fig. 8. IRAS model in a corridor.

TABLE I
CRITICAL PARAMETERS OF IRAS

Parameter	Value
Sound velocity	343 m/s
Room size	3 m × 50 m × 3 m
Temperature	25 °C
Humidity	60%
Reflection coefficient	0.6
Attenuation coefficient	0.368 dB/m
Signal-to-Noise Ratio (SNR)	0 dB

respectively. Furthermore, our proposed method outperformed both, achieving even lower mean and 50th percentile errors of 0.16 and 0.05 m, respectively. These results highlight the exceptional precision the proposed methods.

B. IRAS

In this section, we validate our proposed method through simulation using COMSOL 6.0 to construct the acoustic simulation model and MATLAB 2022b for data analysis. COMSOL is chosen due to its capability to precisely simulate CIR characteristics of large-scale indoor structures using ray acoustics.

The simulation model presented in this article primarily consists of two components: 1) room model construction and 2) received signal generation. The room model is assembled using ray acoustics within COMSOL. As Fig. 8 illustrated, our room model is designed as a long corridor, characterized by a large aspect ratio, as well as high levels of multipath interference. A set of simulation parameters, such as room air temperature, humidity, the reflection coefficient of the room walls, and the relative positions of the sound source and microphone, are configured to accurately emulate the room environment, as depicted in Table I. The receivers are evenly

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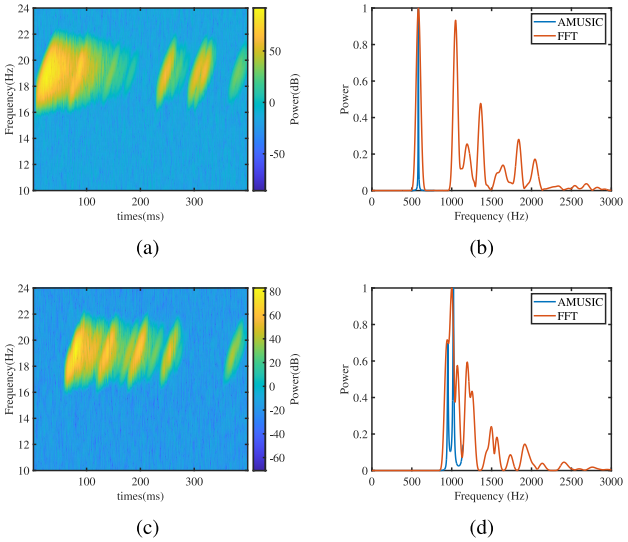


Fig. 9. IRAS spectrum of generated signals and frequency-domain conversion of multipath effects at various distances: (a) spectrum of the received signal at 10 m; (b) frequency spectrum of the mixed signal for (a); (c) spectrum of the received signal at 15 m; and (d) frequency spectrum of the mixed signal for (c).

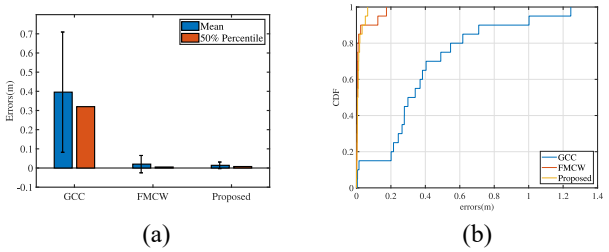


Fig. 10. IRAS simulation comparison results with FFT and AMUSIC. The main purpose of this simulation is to validate the effectiveness of AMUSIC. (a) Comparison bar chart of numerical simulation errors. (b) Numerical simulation CDF diagram.

distributed in the range of 0–20 m. The simulation parameters are comprehensive and critical for the study's accuracy. These parameters include a sound velocity of 343 m/s, room dimensions of 3 m × 50 m × 3 m, a temperature of 25 °C, a humidity level at 60%, a reflection coefficient of 0.6, an attenuation coefficient of 0.294 dB/m, and a signal-to-noise ratio (SNR) set to 0 dB. Each parameter plays a vital role in the fidelity and realism of the acoustic simulation. The obtained CIR is then employed to generate the actual received signal. The simulation assumes uniform sound velocity and smooth wall surfaces.

In this study, the ranging accuracy of our proposed method is rigorously evaluated in a scenario characterized by intense multipath interference, highlighting its superiority over the conventional GCC approach. Following the generation of CIR, actual signals are derived as per (31). Fig. 9 displays the IRAS signals and the FMCW frequency domain. These results indicate that IRAS can effectively simulate multipath effects, and our method successfully translates different arrival paths into distinct and high-resolution frequency components.

Fig. 10 shows the ablation IRAS comparison results between GCC, FFT [19] and the proposed method. It could

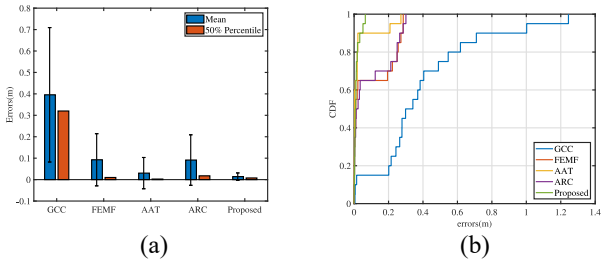


Fig. 11. IRAS simulation comparison results with benchmark methods. (a) Comparison chart of numerical simulation errors. (b) Numerical simulation CDF diagram.

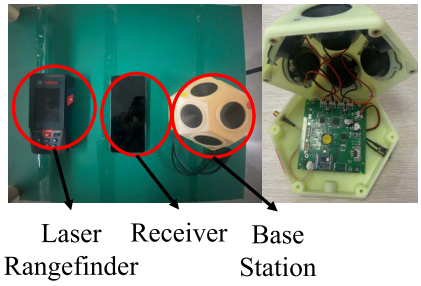
be found that the TOA estimation error in GCC in the dense reverberation environment is large and unstable. In contrast, both the FFT approach and our proposed method achieve higher accuracy. Notably, due to its superior resolution and absence of sidelobe interference, our proposed method secures the highest accuracy with minimal standard error.

Our results are benchmarked against several established methods: GCC [20], [21], a first-extremum match-filter method (FEMF) [26], and two-stage audio arrival time methods (AAT) [13], [16] and an improved TOA estimation method in a reverberate environment (ARC) [24]. As illustrated in Fig. 11, our proposed method outperforms these conventional techniques, demonstrating enhanced performance with a mean error of 1.4 cm.

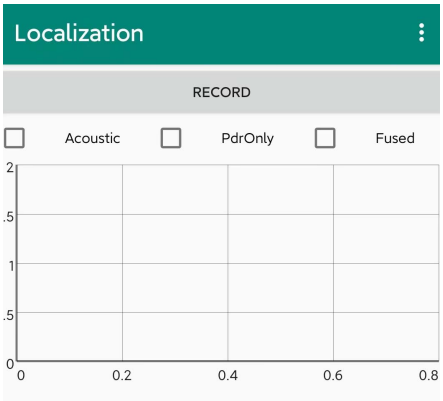
IV. Experiment

In this section, experiments are carried out in an indoor corridor situated on the Yuquan Campus of Zhejiang University to evaluate the real-world performance of the proposed method. The dimensions of the corridor are 2.6 m × 2.7 m × 50 m, whose shape is identical to the IRAS model to correspond with the simulation model. The emitter employed is a custom-developed acoustic base station, while the microphone, is situated at the bottom of the Huawei P20 Pro. The base station uses a low-cost microcontroller STM32F103, and the frequency response range of the speaker is 20 Hz–24 kHz. Fig. 12(a) illustrates the experimental base station and its structure. The received signal is recorded in real-time through the positioning app installed on our smartphone, as shown in Fig. 12(b). Our recent research [34], [35], [36] validated that the self-developed acoustic base station can be used in indoor ranging and localization. To mitigate synchronization interference between the base station and the receiver, the devices are initially positioned nearby at the onset of the ranging experiment, where the separation is 0 m and the quality of the received signal is sufficiently high. This arrangement is crucial to accurately ascertain the actual transmission time of the signal. Following this initial setup, the device is quickly relocated to the designated measurement point for distance assessment. Throughout the experiment, the speaker produces a sound pressure level of 55 dB at a distance of 1 m. The acoustic Chirp signal used in the experiment has a duration of 40 ms and a bandwidth of 4 kHz, from 17 to 21 kHz.

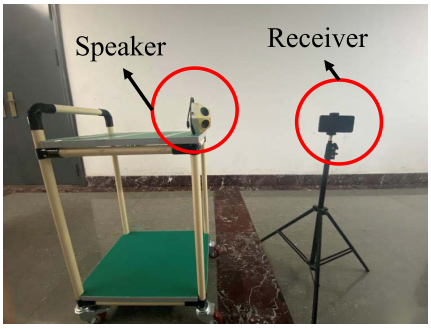
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(a)



(b)



(c)



(d)

Fig. 12. Experiment scenarios and setting. The experimental framework encompasses scenarios in both LoS and NLoS conditions, as well as the corresponding settings and instrumentation. (a) Experimental base station and its internal structure. (b) Experimental smartphone acoustic signal receiver application. (c) Experimental setting in LoS. (d) Experimental setting in NLoS.

Fig. 12(c) and (d) illustrate the experiment setup. The test points are located at distances of 5.328, 10.946, 16.21, 21.603, 25.067, and 29.727 m for LoS. And 10.975, 21.625, and 29.745 m for NLoS. The NLoS setup is designed to replicate typical scenarios where human occlusion is common, thereby simulating realistic environmental conditions. We employ a laser rangefinder to measure the ground truth, achieving a measurement accuracy of millimeter-level. The experiment is repeated 60 times for each point. Each experiment, LoS and NLOS, are included in the four method evaluation mean errors, standard errors, 50th percentile errors and 95th percentile errors. All results are compared with the same benchmark methods in the simulation: GCC [20], [21], FEMF [26], AAT [13], [16], and ARC [24].

Fig. 13 illustrates the ranging accuracy at varying distances from 0 to 30 m, demonstrating the superior performance and robustness of the proposed method in error minimization. The proposed method's effectiveness stems from its ability to transform multipath effects into the frequency domain using AMUSIC, enhancing noise removal and resolution, and thereby facilitating easier TOA estimation. The maximum error in ranging measurements remains below 0.1 m for distances up to 20 m. Beyond this range, errors exceed 0.1 m due to increased multipath interference and signal attenuation. Nonetheless, the proposed method consistently outperforms others in terms of TOA estimation accuracy and robustness across all tested distances. Additionally, the CDF plots indicate that our proposed algorithm achieves the smallest median error with the fewest outliers, which illustrates our robustness in TOA estimation.

Fig. 14 shows the ranging accuracy in NLOS scenarios for distances between 10 and 30 m. The proposed method significantly improves accuracy in these challenging conditions, reducing errors by approximately 0.1 m compared to other methods. Under NLOS conditions, where the reverberation time is typically extended and the direct path is greatly diminished or virtually absent, there is a crucial need for enhanced frequency estimation accuracy. Our methodology, employing FMCW and AMUSIC, excels in accurately estimating frequencies even in NLOS environments. In contrast, the GCC method struggles, as it often yields outputs with multiple overlapping peaks, obscuring the identification of the initial path. The performance improvement of our approach is even more evident when considering the improvement margin, which is nearly 0.2 m in accuracy. This underscores the substantial benefits of the proposed method in NLOS conditions, where it maintains superior accuracy levels compared to other methods, which experience significant performance degradation.

The TOA estimation accuracy with varying distances are illustrated in Fig. 15(a) and (b). As shown in Fig. 15)(a), when the distance is less than 15 m, the errors across different methods are relatively minor. However, as the distance increases, the errors for other comparison methods, particularly GCC, rise significantly. While utilizing the first path and its derived methods can mitigate errors, the method proposed in this article maintains errors at the centimeter level, even at distances up to 25 m, without a significant increase.

As shown in Fig. 15(b), due to the inability to find a direct signal path in these scenarios, the error does not significantly increase with distance. Additionally, methods,

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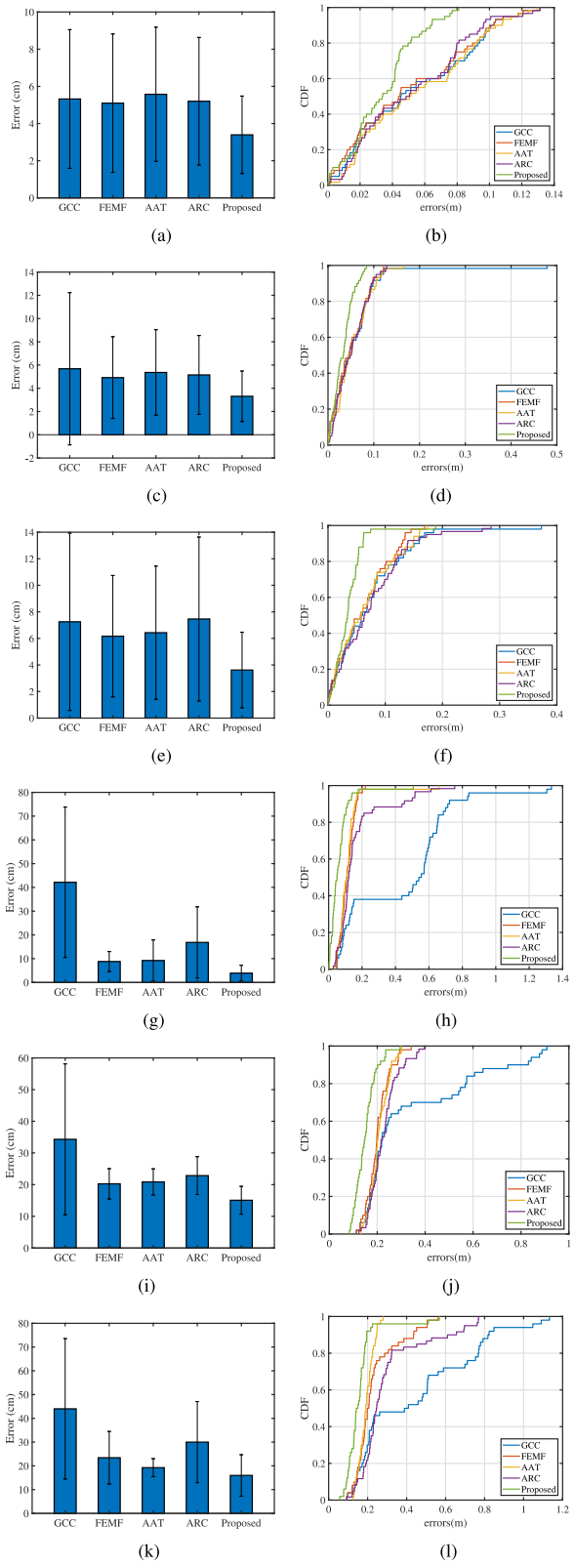


Fig. 13. LOS experiment comparative results in the bar chart and CDF plot at different distances. (a) 5.328 m. (b) 5.328 m CDF. (c) 10.946 m. (d) 10.946 m CDF. (e) 16.21 m. (f) 16.21 m CDF. (g) 21.603 m. (h) 21.603 m CDF. (i) 25.067 m. (j) 25.067 m CDF. (k) 29.727 m. (l) 29.727 m CDF.

such as removing reverberation (ARC), yield better results in NLOS situations. However, our proposed method consistently achieves the lowest error.

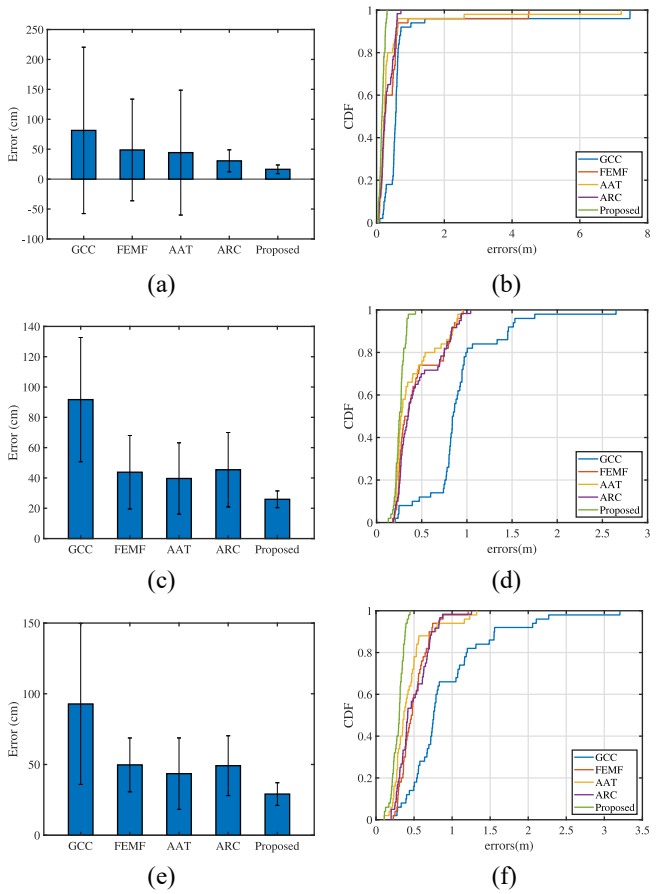


Fig. 14. Accuracy in NLOS experiment results at different distances. (a) NLOS 10.975 m. (b) NLOS 10.975 m CDF. (c) NLOS 21.625 m. (d) NLOS 21.625 m CDF. (e) NLOS 29.754 m. (f) NLOS 29.754 m CDF.

Tables II and III present the detailed experimental results comparing different ranging methods in LOS and NLOS environments. For each method, the table lists the mean error, standard deviation, 50th percentile (median), and 95th percentile of ranging errors across different distances, measured in m. The proposed method shows consistently lower mean errors and standard deviations across all distances compared to the other methods, indicating better performance and reliability. In the LOS environment, particularly at a distance of 21.603 m, the proposed method achieves exemplary results with a mean error of just 0.064 m and a standard deviation of 0.075 m. Furthermore, it maintains a tight error margin, with the maximum mean error and standard deviation recorded as 0.160 and 0.087 m, respectively. This data underscores the robustness and precision of the Proposed method in LoS.

Conversely, Table III highlights the superiority of the Proposed method in terms of ranging accuracy, particularly in NLOS conditions. It consistently outperforms other evaluated methods, delivering lower mean errors and standard deviations in both LOS and NLOS scenarios. In challenging NLOS environments, the Proposed method markedly reduces error magnitudes, thereby improving both accuracy and robustness. The table indicates that the largest mean error and standard deviation for the Proposed method in NLOS conditions are

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TABLE II
ANALYTICAL RESULTS OF THE EXPERIMENT ACROSS VARIOUS DISTANCES, ENVIRONMENTS, AND METHODS IN LOS

Environment	Method	Distance(m)	Mean(m)	Standard Deviation(m)	50th percentile(m)	95th percentile(m)
LOS	GCC	5.328	0.053	0.037	0.045	0.102
	FEMF		0.051	0.037	0.044	0.101
	AAT		0.056	0.036	0.051	0.106
	ARC		0.052	0.034	0.050	0.096
	Proposed		**0.034**	**0.021**	**0.034**	**0.064**
	GCC	10.946	0.060	0.066	0.049	0.102
	FEMF		0.053	0.036	0.047	0.100
	AAT		0.057	0.038	0.051	0.105
	ARC		0.051	0.033	0.049	0.096
	Proposed		**0.033**	**0.021**	**0.033**	**0.065**
	GCC	16.210	0.073	0.067	0.059	0.155
	FEMF		0.062	0.046	0.057	0.130
	AAT		0.064	0.050	0.058	0.149
	ARC		0.074	0.061	0.064	0.139
	Proposed		**0.036**	**0.028**	**0.034**	**0.062**
	GCC	21.603	0.445	0.316	0.523	0.694
	FEMF		0.111	0.042	0.085	0.142
	AAT		0.115	0.087	0.084	0.135
	ARC		0.168	0.150	0.120	0.446
	Proposed		**0.064**	**0.075**	**0.049**	**0.115**
	GCC	25.067	0.343	0.238	0.219	0.788
	FEMF		0.203	0.048	0.198	0.272
	AAT		0.209	0.041	0.202	0.259
	ARC		0.228	0.059	0.219	0.315
	Proposed		**0.151**	**0.044**	**0.147**	**0.208**
	GCC	29.727	0.440	0.296	0.399	0.820
	FEMF		0.234	0.111	0.202	0.436
	AAT		0.192	0.038	0.191	0.246
	ARC		0.299	0.170	0.246	0.632
	Proposed		**0.160**	**0.087**	**0.146**	**0.195**

TABLE III
ANALYTICAL RESULTS OF THE EXPERIMENT ACROSS VARIOUS DISTANCES, ENVIRONMENTS, AND METHODS IN NLOS

Environment	Method	Distance(m)	Mean(m)	Standard Deviation(m)	50th percentile(m)	95th percentile(m)
NLOS	GCC	10.975	0.814	1.391	0.563	0.714
	FEMF		0.487	0.850	0.232	0.611
	AAT		0.443	1.044	0.231	0.597
	ARC		0.305	0.184	0.244	0.575
	Proposed		**0.164**	**0.072**	**0.160**	**0.260**
	GCC	21.625	0.917	0.410	0.845	1.456
	FEMF		0.438	0.242	0.333	0.831
	AAT		0.396	0.236	0.272	0.842
	ARC		0.454	0.246	0.349	0.830
	Proposed		**0.259**	**0.055**	**0.254**	**0.333**
	GCC	29.754	0.928	0.569	0.758	1.555
	FEMF		0.497	0.190	0.471	0.717
	AAT		0.435	0.252	0.367	0.753
	ARC		0.491	0.211	0.417	0.752
	Proposed		**0.291**	**0.080**	**0.306**	**0.385**

0.291 and 0.080 m, respectively. This represents a substantial enhancement in performance, with the proposed method improving by nearly 0.5 m in mean error and 0.2 m in standard deviation when compared to alternative methods. This data emphasizes the Proposed method's significant advancements in precision and reliability for ranging measurements under various conditions.

Furthermore, Tables II and III also highlight that the proposed method ensures a more reliable error distribution, as evidenced by the 95 percentile values. In LoS conditions, the highest 95th percentile value recorded for the proposed method is 0.208 m, which is considerably higher in NLoS conditions, peaking at just 0.385 m. This contrasts markedly with the nearly 0.8-m 95th percentile values observed for the other methods compared in the study, confirming the proposed method's enhanced reliability and precision over a diverse set of ranging conditions.

To clarify the critical threshold of λ as defined in (15) and depicted in Fig. 16(a), our analysis demonstrates that a λ value of 0.2 optimally reduces errors consistently across various distances. Notably, for distances less than 10 m, λ has minimal impact. However, as the distance increases beyond this threshold, the influence of λ on error reduction becomes more significant, highlighting its importance in longer range scenarios.

Similarly, Fig. 16(b) illustrates the threshold of λ_2 in (28) when $\lambda = 0.2$. It is observed that for distances greater than 15 m, λ_2 values exceeding 0.2 lead to a gradual decrease in

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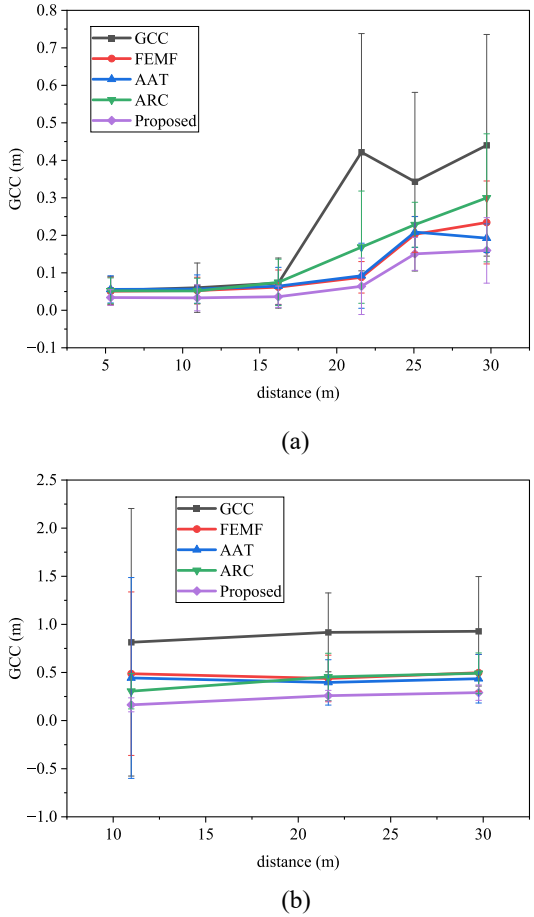


Fig. 15. Performance of TOA measurements in a noisy and reverberant corridor environment. (a) Performance of experiments in LoS at varying distances. (b) Performance of experiments in NLoS at varying distances.

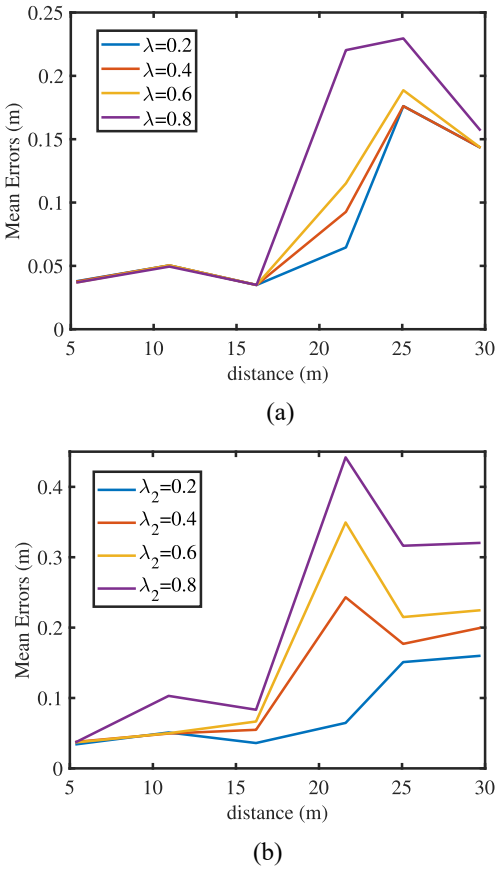


Fig. 16. TOA estimation performance with different λ, λ_2. (a) Error variability across diverse ranging distances as a function of λ. (b) Error variability across diverse ranging distances as a function of λ_2 when $\lambda = 0.2$.

performance, particularly beyond 20 m. Therefore, this study sets the optimal values of both λ and λ_2 to 0.2.

V. Conclusion

In this article, we introduce FDME-RATE, a novel method for high-accuracy TOA estimation in indoor acoustic localization. Departing from traditional GCC-based methods, FDME-RATE utilizes an advanced FMCW approach, effectively transforming multipath signals into the frequency domain. Additionally, the AMUSIC technique is employed to dynamically adjust MUSIC parameters, enhancing ranging accuracy and resolution in both LoS and NLoS conditions.

The effectiveness of FDME-RATE was validated through numerical simulations and IRAS, replicating practical conditions. Comparative analyses, both self-referential and against existing methods, demonstrate the superior performance of our approach. Furthermore, real-world experiments in LoS and NLoS environments yielded an average TOA ranging error of 0.16 m over 30 m in LoS and 0.25 m in NLoS, outperforming previously reported results in the literature.

Despite its contributions, our study is not without limitations, notably the extensive time required for analysis and the lack of experiments in acoustic localization. Future research will concentrate on incorporating acoustic localization experiments and enhancing efficiency to reduce time expenditure. Additionally, it will be critical to validate the applicability of our findings across multiple mobile phone brands to ensure broader applicability and robustness of the results.

Acknowledgment

This article is an extended conference paper of the 2023 13th International Conference on Indoor Positioning and Indoor Navigation (IPIN) N. Jia, W. Cui, and Z. Wang, “Robust acoustic TOA estimation based on multipath extraction in frequency domain,” in 2023 13th IPIN, pp. 1–7, 2023.

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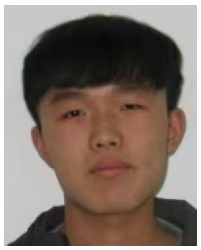
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